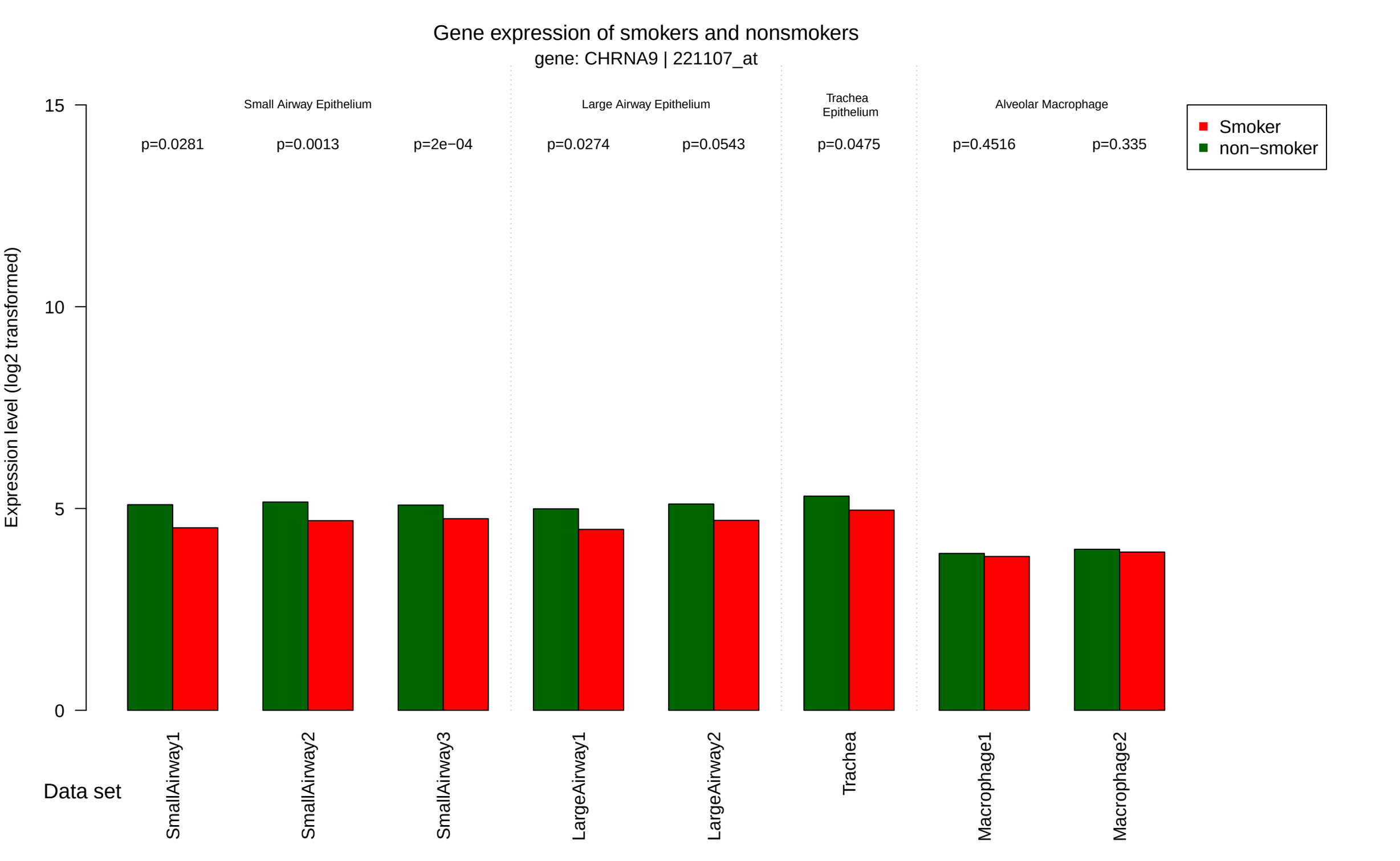
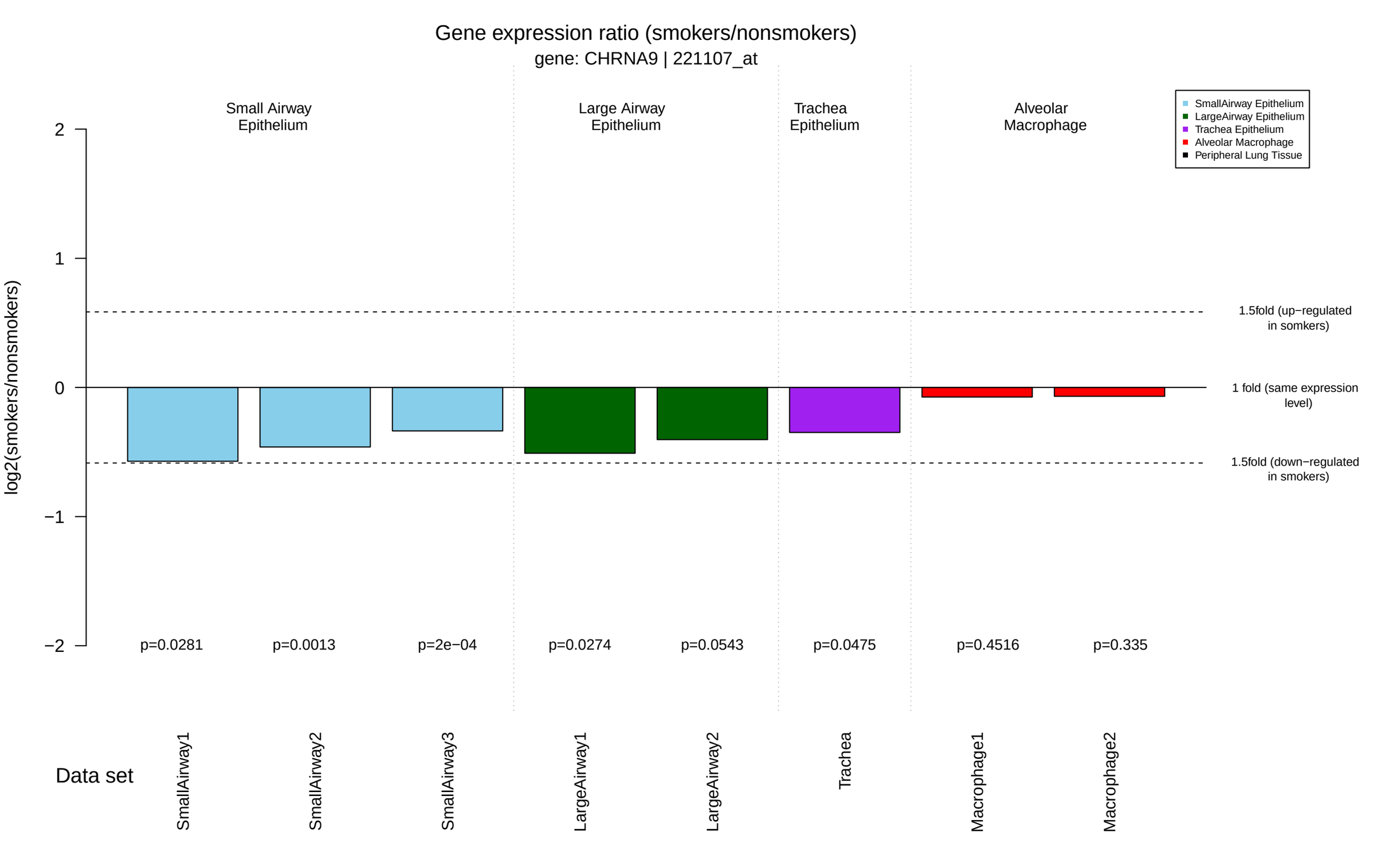
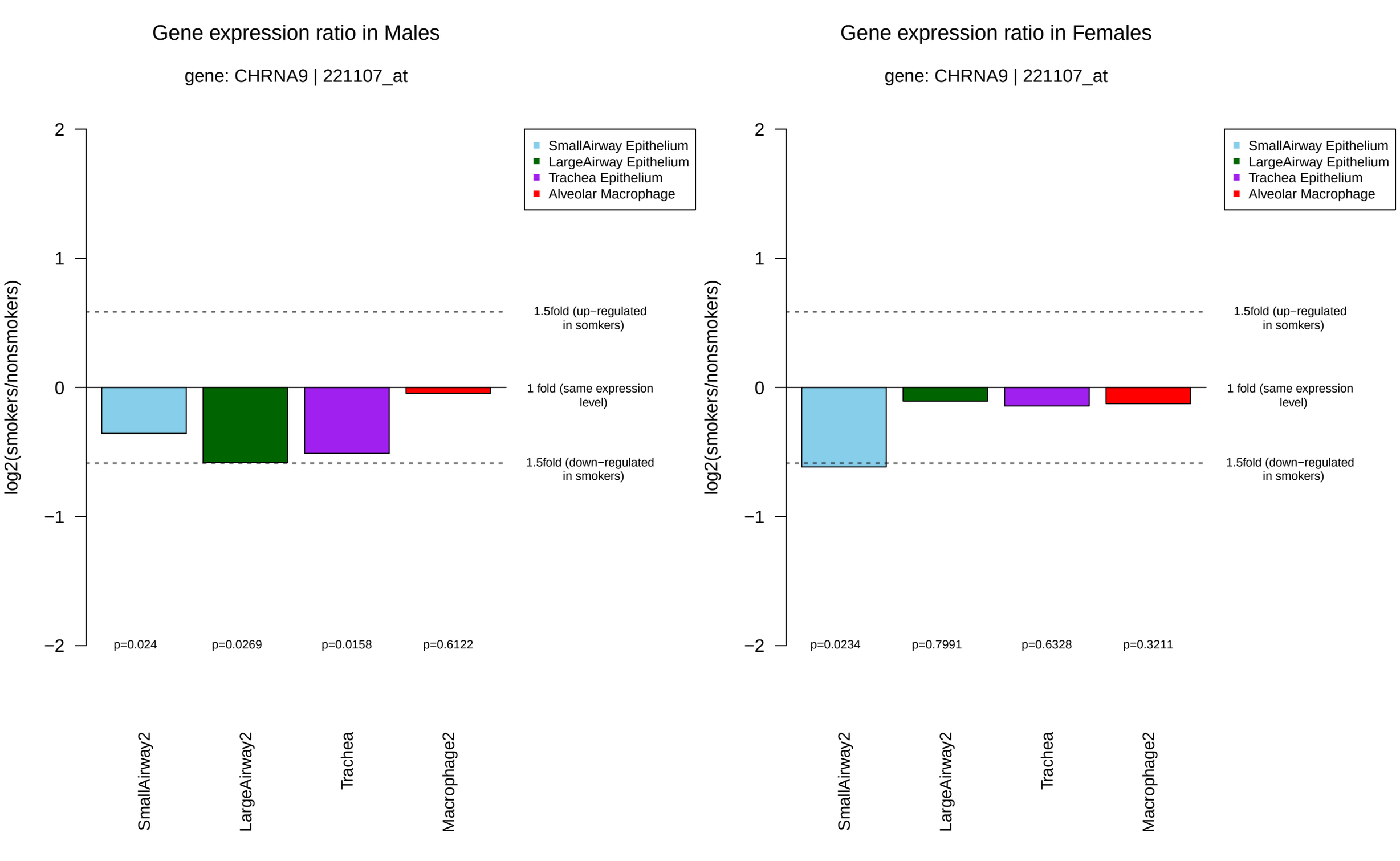
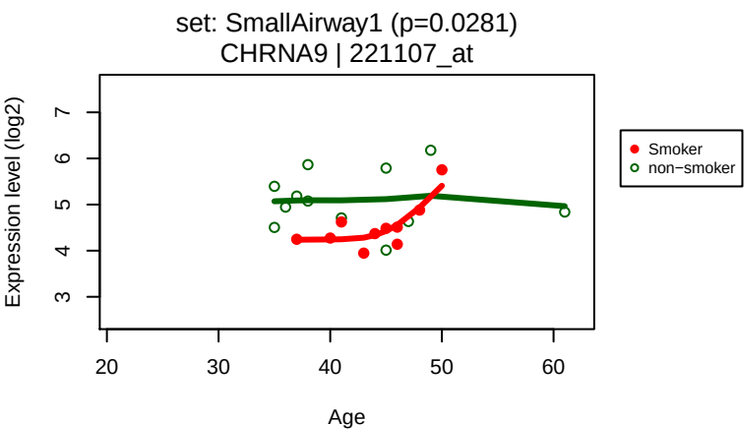
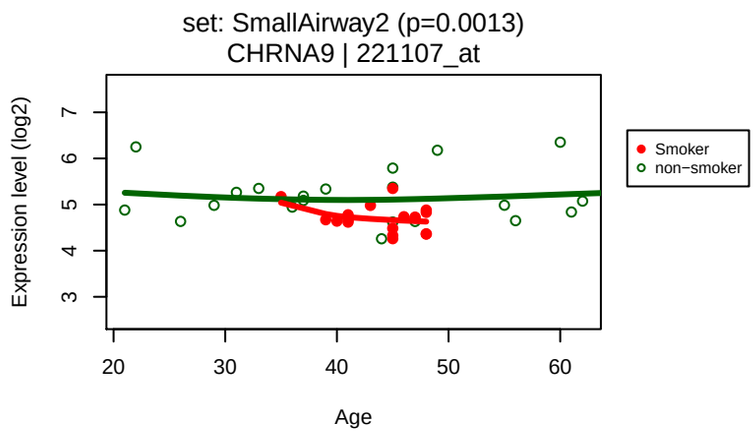
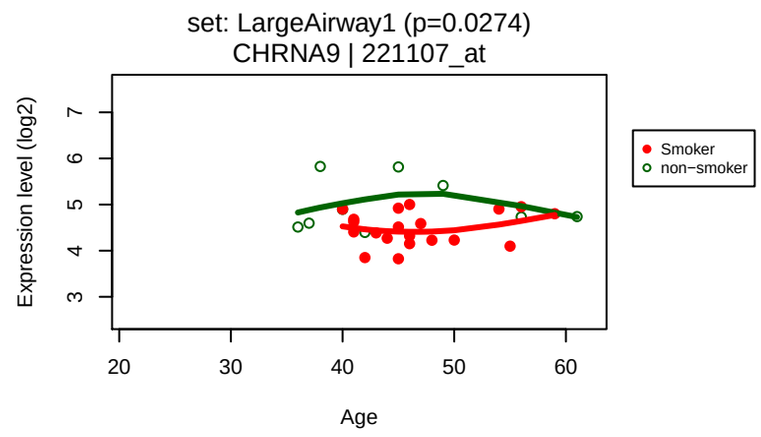

Gene expression of smokers and nonsmokers
gene: CHRNA9 | 221107_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0281
p=0.0013
p=2e−04
p=0.0274
p=0.0543
p=0.0475
p=0.4516
p=0.335
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

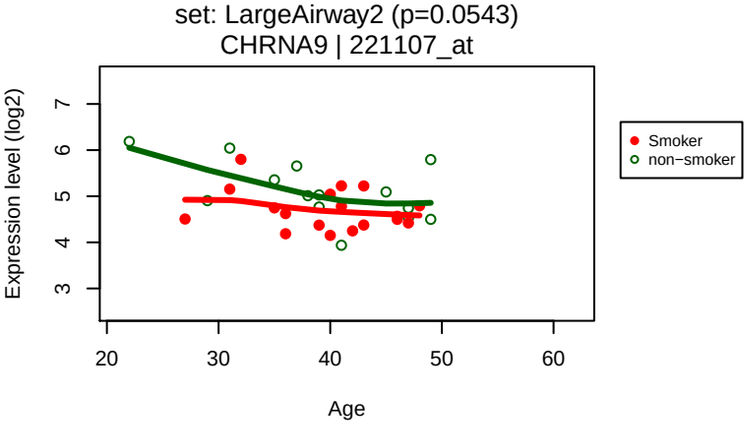

Gene expression ratio (smokers/nonsmokers)
gene: CHRNA9 | 221107_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0281
p=0.0013
p=2e−04
p=0.0274
p=0.0543
p=0.0475
p=0.4516
p=0.335
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

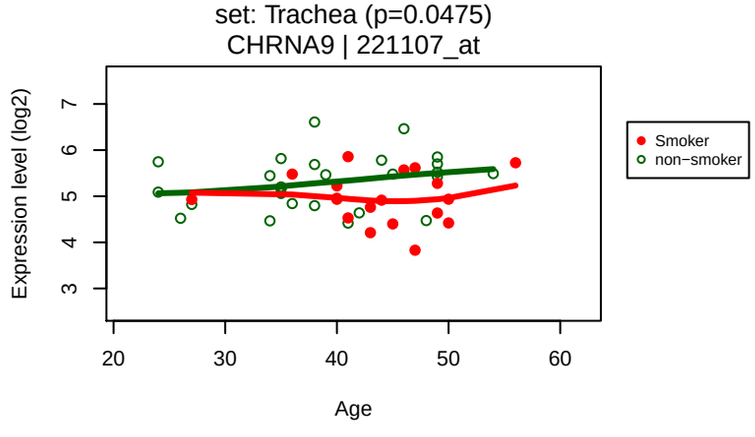

Gene expression ratio in Males
gene: CHRNA9 | 221107_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.024
p=0.0269
p=0.0158
p=0.6122
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CHRNA9 | 221107_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0234
p=0.7991
p=0.6328
p=0.3211
SmallAirway2
LargeAirway2
Trachea
Macrophage2

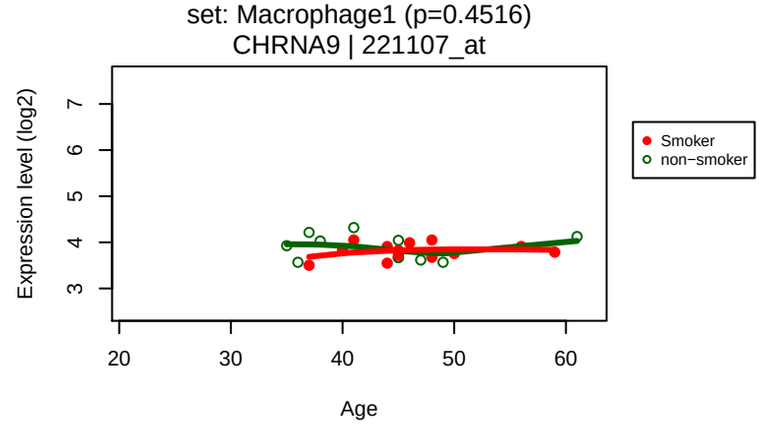

set: SmallAirway1 (p=0.0281)
CHRNA9 | 221107_at
Expression level (log2)
Age
Smoker
non−smoker

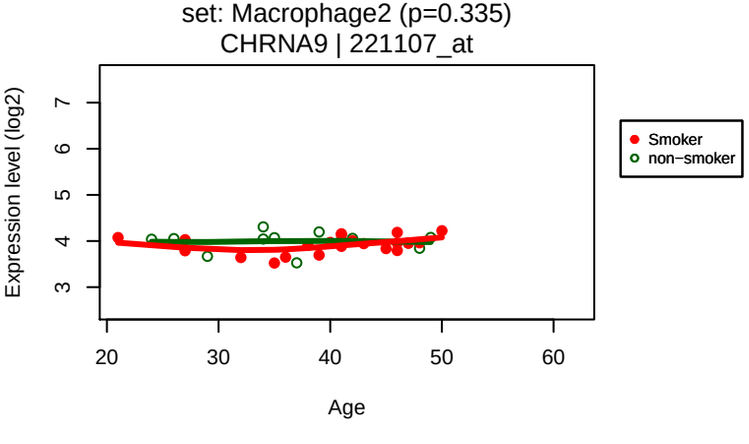

set: SmallAirway2 (p=0.0013)
CHRNA9 | 221107_at
Expression level (log2)
Age
Smoker
non−smoker

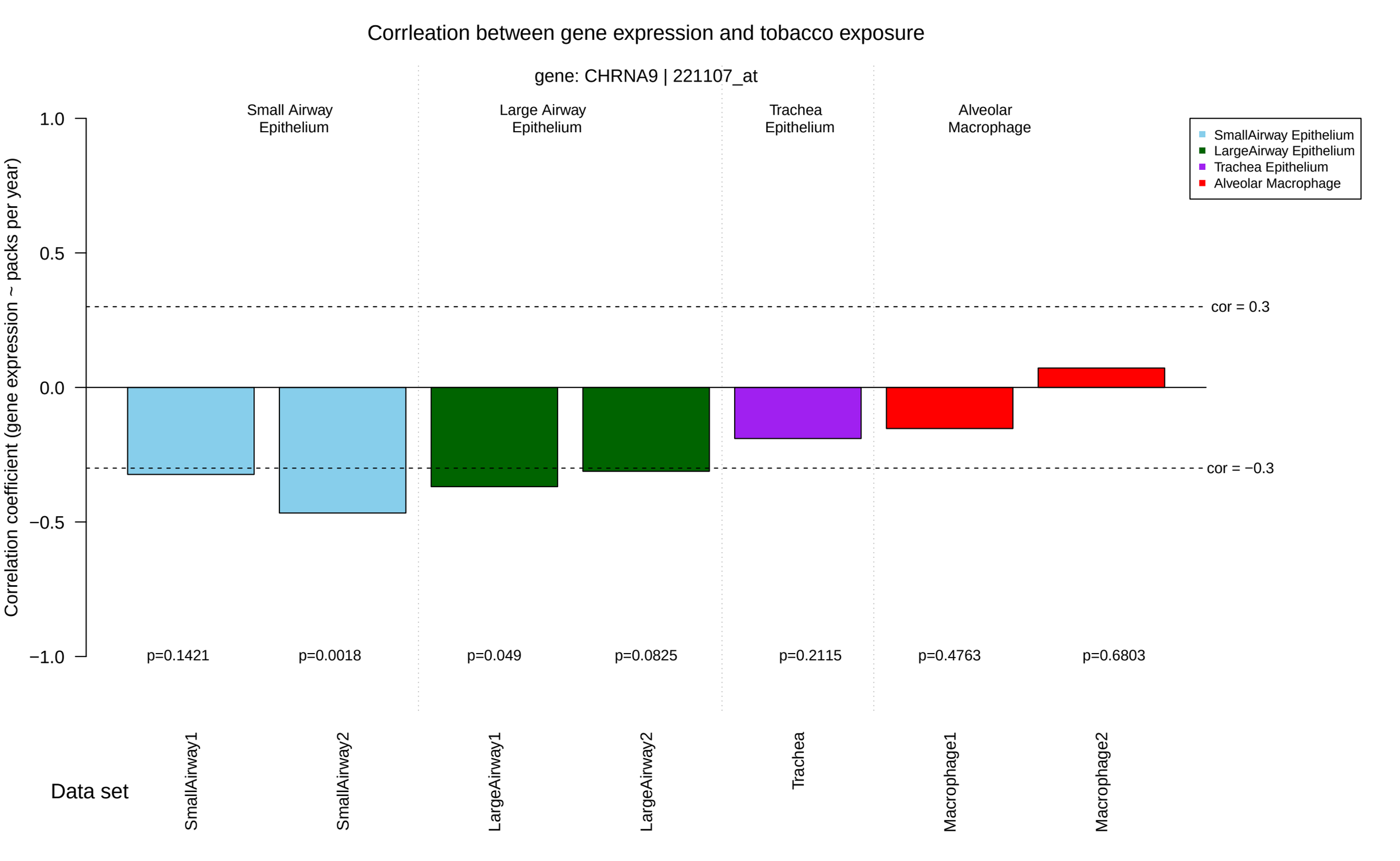

set: LargeAirway1 (p=0.0274)
CHRNA9 | 221107_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0543)
CHRNA9 | 221107_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0475)
CHRNA9 | 221107_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4516)
CHRNA9 | 221107_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.335)
CHRNA9 | 221107_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CHRNA9 | 221107_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1421
p=0.0018
p=0.049
p=0.0825
p=0.2115
p=0.4763
p=0.6803
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2